

Building a Godot Plugin with GDExtension

GodotCon Boston 2026





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Who am I?

- Games and tools developer
- Worked with Unity, Unreal and Godot
- Built a lot of what I work on as open source projects
- Currently building my own game engine (in SDL)
- And a multi-engine tool for building rhythm games (Unity, Unreal, Godot, MonoGame, SDL, etc)
- A cat dad (most important).



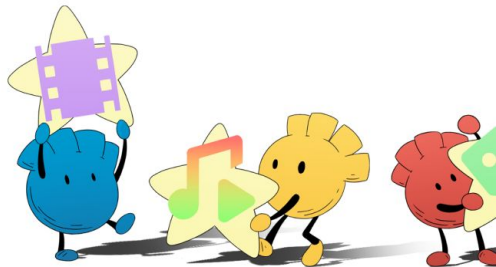
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Outline of Workshop

1. Prerequisites (macOS / Windows / Linux)
2. Setup the repo
3. Building with scons for the first time
4. Creating an empty plugin
5. Loading a simple header library in the plugin
6. Extending a node with custom logic
7. Getters and setters
8. Signals
9. Convert data
10. Migrating a GDScript function to C++
11. Building and Signing (macOS) via GitHub Actions

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Prerequisites

(macOS / Windows / Linux)

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Prerequisites

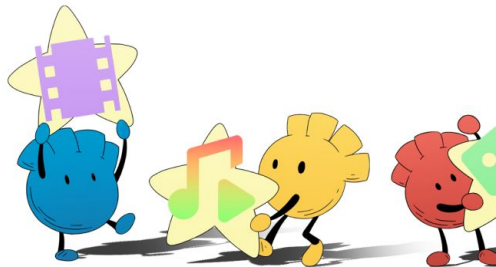
- Godot 4.4+
- VS Code (or equivalent text editor)
- Git
- Python
- Scons

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Prerequisites (macOS)

- Git (<https://git-scm.com/>)
- Brew (<https://brew.sh/>)
- Python - `brew install python`
- Scons - `brew install scons`



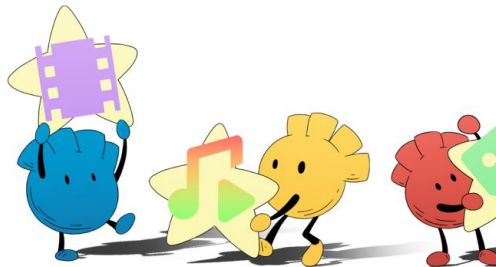
Prerequisites (Windows)

- Git (<https://git-scm.com/>)
- Visual Studio (<https://visualstudio.microsoft.com/vs/community/>) - With Desktop development with C++
- Python (<https://www.python.org/>) - Check off add to ENV variable
- Scons - `python -m pip install scons`



Prerequisites (Linux)

- git - `sudo apt install git`
- python - `sudo apt install python3`
- scons - `sudo apt install scons`
- vim - `sudo apt install vim`



Setup the repo

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Setup the repo

Remember: Commit after each step

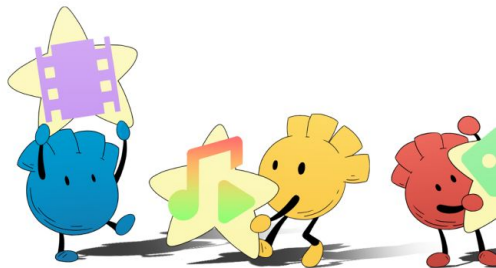
1. Create a new directory and set up a git repo with `git init`
2. Create a README.md file
3. Add the **godotengine/godot-cpp** repo as a submodule

```
git submodule add https://github.com/godotengine/godot-cpp.git
```

- a. Navigate into the godot-cpp/ directory
- b. Checkout the tag for the minimum version of Godot you want to support

```
git checkout godot-4.4-stable
```

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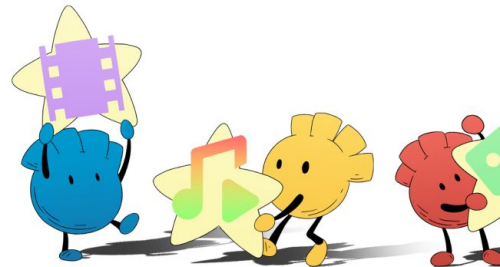
Building with scons for the first time

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Creating an empty plugin

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Loading a simple header library in the plugin

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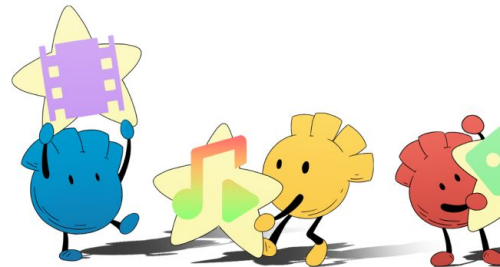
Extending a node with custom logic

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Getters and setters

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Signals

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Convert data

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Migrating GDScript function to C++

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Building and Signing (macOS) via GitHub Actions

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Questions?

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